

Robust Pose-Sequence Retrieval with Contextual Metric Learning

Final-proposal discussion brief

July 2026

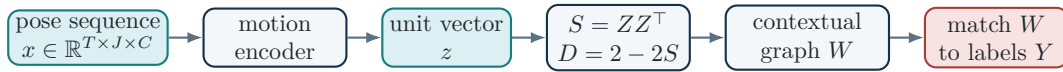
60-second pitch

I propose to transfer contextual metric learning from image retrieval to temporal pose embeddings. A pretrained motion encoder maps each skeleton sequence to a normalized vector, and cosine similarity ranks a clean reference gallery. The experimental question is whether optimizing *neighborhood structure*, rather than only direct pairs, causes retrieval quality to degrade more slowly when query poses contain joint jitter, missing joints or limbs, dropped frames, or pose-estimator shift. I would hold the encoder, split, sampler, augmentations, and evaluator fixed; compare supervised contrastive, contextual, and hybrid objectives; and report Recall@1/5/10, mAP, and clean-to-corrupted degradation.

Scientific boundary

The foundation paper establishes reduced overfitting and label-noise robustness for *image retrieval*. It does not prove robustness to corrupted pose coordinates. That transfer is the hypothesis to test, not a premise to assume.

Method in one picture



Five equations to narrate

$$\begin{aligned}
 D(i, j) &= \|z_i - z_j\|_2^2 = 2 - 2s_{ij}. \\
 N_{ij} &= \theta(-D(i, j) + \text{sg}(D(i, p_k)) + \epsilon), \quad \frac{\partial \theta}{\partial x} = \alpha. \\
 W_1(i, j) &= \frac{N_{ij}}{2} \left[\frac{M_+(i, j)}{\text{sg} \sum_p N_{ip}} + \frac{M_-(i, j)}{\text{sg} \sum_p (1 - N_{ip})} \right]. \\
 W_2 &= \frac{RW_1}{R\mathbf{1}}, \quad W = \frac{1}{2}(W_2 + W_2^\top), \quad R = N_{k/2} \odot N_{k/2}^\top. \\
 \mathcal{L} &= \lambda \frac{1}{n^2} \sum_{i \neq j} (y_{ij} - w_{ij})^2 + (1 - \lambda) \mathcal{L}_{\text{contrast}} + \gamma \mathcal{L}_{\text{reg}}.
 \end{aligned}$$

The key theoretical statement

For a balanced batch with exactly k samples per class, $n > 2k$, and $\epsilon = 0$, zero contextual loss occurs exactly when every positive ranks ahead of every negative within the batch. This is why k is tied to samples per class rather than treated as an unrelated top- k hyperparameter.

Experimental design and discussion answers

Controlled experiment

Component	Decision
Dataset	NTU RGB+D 120; official cross-subject and cross-setup protocols.
Encoder	Pretrained MotionBERT sequence encoder; normalized projection output.
Retrieval	Disjoint query/gallery manifest; noisy queries against a fixed clean gallery.
Losses	Cross-entropy embedding, SupCon, triplet or multi-similarity, contextual, contextual+contrastive, corruption-aware SupCon.
Corruptions	Gaussian joint jitter; random joint/limb masking; frame dropout; 3D yaw rotation; pose-estimator shift when available.
Metrics	Recall@1/5/10, mAP, retention $M(c)/M(0)$, and degradation $\Delta(c) = M(0) - M(c)$.
Fairness	Same backbone, embedding dimension, sampler, split, tuning budget, augmentations, and seeds.

Lowest-risk implementation sequence

1. Freeze MotionBERT; validate splits, cosine retrieval, metrics, and deterministic corruptions.
2. Train only small projection heads with SupCon, contextual, and hybrid losses.
3. Profile contextual $O(n^3)$ batch operations; use balanced batches such as $16 \times 4 = 64$ or $32 \times 4 = 128$.
4. Fine-tune the full encoder only after the loss-side comparison is stable.
5. Add corruption-aware training and a MaskCLR-style robustness control.

Likely questions and precise answers

Question	Answer
Why retrieval rather than classification?	Classification can remain acceptable while the ordering of the embedding space deteriorates. Retrieval measures whether examples remain inspectable and useful for one-shot decisions.
Where is the novelty?	Not sequence retrieval itself. The bounded gap is whether the contextual objective improves <i>retrieval degradation curves under input corruption</i> when the temporal encoder and evaluator are fixed.
Why might neighborhoods help?	A direct similarity can be unstable under local pose noise; shared and reciprocal neighborhoods provide a local-consensus signal. This is an inductive-bias hypothesis, not a guarantee.
Why NTU120 first?	It provides scale, many examples per class, standard protocols, and compatibility with existing skeleton and one-shot pipelines. It supports a reproducible first test, not claims about fine-grained technique quality.
What would a null result mean?	Robustness is mainly attributable to the encoder or corruption exposure rather than the contextual objective. That is still an informative controlled result.
What is the main engineering risk?	Cubic batch computation and batch-structure sensitivity. Both are bounded by moderate balanced batches, matrix operations, and staged profiling.

Do not claim

Do not claim that action-class retrieval measures movement quality, that label-noise robustness implies pose-noise robustness, or that a new motion architecture is being proposed. The contribution is a fair retrieval-first robustness test of the loss.

Closing sentence. *The project is deliberately narrow: it asks whether local neighborhood structure is a useful inductive bias when the observations, rather than the labels, become unreliable.*